TREY LUPO

AI/ML Engineer

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SUMMARY

AI/ML engineer who ships production systems, not notebooks — a statistically rigorous AutoML framework in **Rust**, Python SDKs that automate Oracle EPM workflows, and ML pipelines deployed inside a corporate finance org. Pairs real modeling discipline (calibration, conformal prediction, temporal validation) with full-stack delivery (Python/FastAPI, React/TypeScript, Claude API). B.A. Statistics, MIT IDSS Data Science certificate. Languages: Python · Rust · TypeScript · SQL · R.

EXPERIENCE

Franklin Street — FP&A Analyst (ML & AI lead)

Tampa, FL · Sep 2025 – Present

- Shipped a production **ML deal-closure prediction pipeline**: AUC **0.874**, calibration error 4.8%; ExtraTrees + XGBoost with **MAPIE conformal prediction**, walk-forward temporal validation, 51 automated tests.
- Built full-stack **AI financial dashboard** (React 19/TypeScript + FastAPI + Azure SQL) with **AI variance narratives via Claude API streaming**, consolidated P&L, drill-downs, three-tier caching — replacing a **132-page monthly report** and cutting variance analysis from **8+ hours to under 1 hour**.
- Built **Oracle NSPB / EPM Cloud automation** — production ETL for revenue and valuation-model uploads with **114+ tests** (open-sourced as the *nspb-api* SDK).
- Led enterprise **AI adoption** as "AI Champion" — scaled Claude from **5 to 15 users** with onboarding curriculum, shared prompt library, and licensing business case.
- Designed **KERNEL Method v3.0**, an 8-stage framework for transparent formula-driven financial narratives used across all **7 business lines**.
- Authored a **12-market CRE analysis workbook** (37 sheets, 2,000+ formulas, 63 charts) informing decisions across a **$500M+ portfolio**; caught a ~$270B multifamily data gap that fed the **CFO's board presentation**.

Origin Construction — AI Engineering Intern

Jun 2025 – Sep 2025

- APM automation, Procore integrations, and AI agent pilots for construction project management workflows.

SELECTED PROJECTS full set at treylupo.com

- **FerroML** — statistically rigorous AutoML framework in **Rust** with Python bindings; **55+ algorithms**, **6,500+ tests**, on **PyPI**; hypothesis-driven model selection with proper statistical testing.
- **nspb-api** — complete Python SDK for Oracle EPM Cloud (Planning, Migration, Data Management); **60+ API endpoints**.
- **Hierarchical Revenue Forecast / DealForecast** — multi-LOB forecasting with hierarchical time series, conformal intervals, SHAP explainability (AUC **0.874**).
- **KKBOX Churn Prediction** — leak-safe features, isotonic calibration, temporal backtests, PSI drift monitoring on a **30GB+** user transaction dataset; **0.97 AUC**; live on Hugging Face Spaces.
- **Open-Vocab Object Detection** — text-prompted detection + segmentation (Grounding DINO + SAM 2), 265–490 ms/image on RTX 3070.
- **AI tooling** — MLForge & GSD-ML (autonomous tabular ML research) and a FerroML Claude Code plugin.

EDUCATION

- **B.A. Statistics**, University of South Florida — May 2024
- **MIT IDSS** — Data Science & Machine Learning Certificate

SKILLS

Languages: Python · Rust · TypeScript · SQL · R

ML / Data: scikit-learn · XGBoost · LightGBM · PyTorch · Pandas · FastAPI · MAPIE · SHAP

AI tooling: Claude API / Anthropic SDK · Claude Code

Finance: Oracle EPM / NSPB Cloud · NetSuite · FP&A · financial modeling

Web / infra: React · GSAP · Vite · Azure SQL